

Skills Summary

Right Triangles: Choosing the Method

Use this reference while completing your worksheet or when studying.

Read the givens, then pick the tool. Two *sides* and no angle $\rightarrow$ Pythagoras. An angle of 45° , 30° or $60^\circ \rightarrow$ the exact special-triangle ratios. Any other angle, or an unknown angle $\rightarrow$ sine, cosine or tangent.

Skill 1 — The Pythagorean theorem

Rule. With legs a , b and hypotenuse c : $a^2 + b^2 = c^2$. Solving for the hypotenuse you *add* the squares; solving for a leg you *subtract* the known leg's square from the hypotenuse's.

Example. A right triangle has legs 6 and 8. Find the hypotenuse.

Step 1. Two sides are known and no angle is given, so trigonometry has nothing to work with — Pythagoras is the tool, and the unknown is the hypotenuse, so add.

Step 2. $c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$
 $c = 10$

Try it. A right triangle has hypotenuse 25 and one leg 7. Find the other leg.

check: 24

Skill 2 — Special right triangles

Two exact shapes worth memorising. In a 45–45–90 triangle the legs are equal and the hypotenuse is a leg times $\sqrt{2}$. In a 30–60–90 triangle the side opposite 30° is half the hypotenuse, and the side opposite 60° is that short leg times $\sqrt{3}$.

Example. An isosceles right triangle has legs of 5. Find the hypotenuse.

Step 1. Equal legs means the acute angles are both 45° , so the exact 45–45–90 ratio applies and no calculator work is needed to set it up.

Step 2. $c = 5\sqrt{2} = 7.0710\dots$
 $c \approx 7.07$

Try it. An isosceles right triangle has legs of 9. Find the hypotenuse.

check: 12.73

Skill 3 — A side from an angle

Rule. Label the sides relative to the known angle, then choose: $\sin = \text{opp/hyp}$, $\cos = \text{adj/hyp}$, $\tan = \text{opp/adj}$. If the unknown is on top, multiply; if it is underneath, divide.

Example. A right triangle has hypotenuse 20 and one acute angle 28° . Find the side adjacent to that angle.

Step 1. The angle is not one of the special ones, so trigonometry it is; adjacent with hypotenuse is cosine, and the unknown is on top.

Step 2. $\text{adj} = 20 \cos 28^\circ = 17.6589 \dots$
 $\text{adj} \approx \mathbf{17.66}$

Try it. A right triangle has a side of 14 adjacent to a 61° angle. Find the side opposite that angle.

check: 25.25

Skill 4 — An angle from two sides

Rule. Form the ratio the two known sides make, then undo it with the matching inverse: $\text{opp and hyp} \rightarrow \sin^{-1}$; $\text{adj and hyp} \rightarrow \cos^{-1}$; $\text{opp and adj} \rightarrow \tan^{-1}$.

Example. A right triangle has a side of 9 opposite $\angle A$ and a hypotenuse of 20. Find $m\angle A$.

Step 1. An angle is unknown and two sides are known — opposite with hypotenuse is the sine pairing, so the inverse sine recovers the angle.

Step 2. $A = \sin^{-1}(9/20) = 26.7436 \dots^\circ$
 $m\angle A \approx \mathbf{26.74}$ degrees

Try it. A right triangle has legs 5 and 12. Find the acute angle opposite the leg of length 5.

check: 22.62 degrees